2025-2026
MATH

PAPER 2

DELIA MEMORIAL SCHOOL (GLEE PATH)

FIRST TERM EXAMINATION 2025-2026

S3 MATHEMATICS

PAPER 2

Time allowed: 1 hour

Group No.	
Name	
Class	**S3 ( )**
Class No.	**( )**
Date	**12 / 01 / 2026**

INSTRUCTIONS

1. When told to open this book, you should check that all the questions are there. Look for the words '**End of Paper**' after the last question.

2. All questions carry equal marks.

3. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber.

4. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

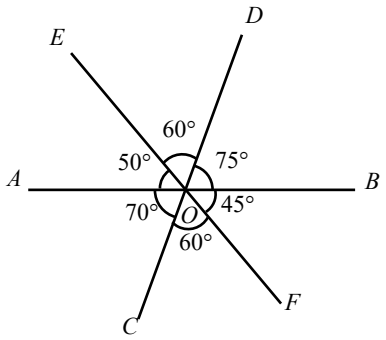
5. No marks will be deducted for wrong answers.

Paper 2		
Total		36

There are 24 questions in Section A and 12 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

1. Refer to the figure.



Which of the following must be true?

I. *AOB* is a straight line.
II. *COD* is a straight line.
III. *EOF* is a straight line.

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

2. $3a^2 - a - 24 =$

A. $(a + 3)(3a + 8)$.
B. $(a - 3)(3a + 8)$.
C. $(a - 3)(3a - 8)$.
D. $(a + 3)(3a - 8)$.

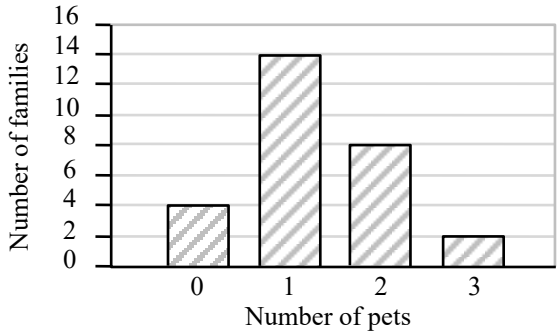
3. $\frac{2^{36}}{2^{-6}} =$

A. 2^{42}.
B. 2^{30}.
C. 2^{6}.
D. 2^{-6}.

4. $-9.023 \times 10^{-7} =$

A. –90 230 000.
B. –9 023 000.
C. –0.000 009 023.
D. –0.000 000 902 3.

5. The bar chart below shows the distribution of the numbers of pets owned by a group of families.



Find the probability that a randomly selected family from the group owns 2 pets.

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{4}{15}$

D. $\frac{1}{3}$

6. Boris goes to school by bus or by MTR. The probabilities that he goes to school by bus and by MTR are 0.25 and 0.75 respectively. The bus fare and the MTR fare are \$10 and \$6 respectively. Find the expected fare for him to go to school.
   A. \$7
   B. \$7.5
   C. \$8
   D. \$9

7. A coffee shop served 500 customers on Friday. The number of customers increased by 6% on Saturday and then decreased by 10% on Sunday. Find the number of customers on Sunday.
   A. 423
   B. 477
   C. 517
   D. 583

8. A certain kind of bacteria grows at a constant rate of 7% per hour. If there are 12 000 bacteria originally, find the number of bacteria after one day, correct to nearest integer.
   A. 12 869
   B. 27 026
   C. 32 160
   D. 60 868

9. The value of a computer has decreased by 12% each year. If the present value of the computer is \$7000, find the value of the computer two years ago.
   A. \$5420, *cor. to 3 sig. fig.*
   B. \$5580, *cor. to 3 sig. fig.*
   C. \$8780, *cor. to 3 sig. fig.*
   D. \$9040, *cor. to 3 sig. fig.*

10. Kathy borrows $\$P$ from a bank at an interest rate of 5% per annum. If she has to pay simple interest of \$1120 after 4 years, then $P =$
    A. 224.
    B. 2800.
    C. 5600.
    D. 6400.

11. The following shows four sums of money are deposited in 4 different banks for the same period. Which option gives the greatest interest?

	Bank	Principal	Simple Interest rate
A.	Bank *A*	\$50 000	0.2% per annum
B.	Bank *B*	\$17 000	0.5% per annum
C.	Bank *C*	\$9600	1% per annum
D.	Bank *D*	\$3800	2% per annum

12. A sum of \$90 000 is deposited at an interest rate of 3.2% per annum for 3 years, compounded quarterly. Find the amount correct to the nearest dollar.
    A. \$99 030
    B. \$98 993
    C. \$98 919
    D. \$98 640

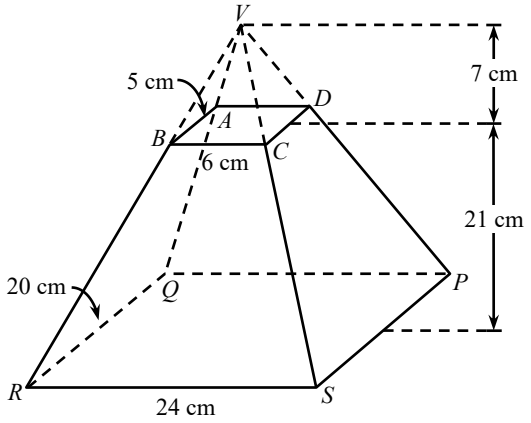
13. The net chargeable income of Helen is \$240 000. Find her salaries tax payable.

	Net chargeable income	**Rate**
On the First	\$50 000	2%
On the Next	\$50 000	6%
On the Next	\$50 000	10%
On the Next	\$50 000	14%
Remainder		17%

    A. \$15 800
    B. \$22 800
    C. \$24 800
    D. \$28 800

14. The base of a solid right pyramid is a square of side 15 cm. If the volume of the pyramid is 825 cm^3, then the height of the pyramid is
    A. 11 cm.
    B. 15 cm.
    C. 33 cm.
    D. 45 cm.

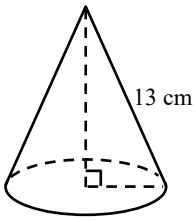
15. The figure shows a frustum *ABCDPQRS* of height 21 cm. The upper base of the frustum is a rectangle of length 6 cm and width 5 cm while the lower base of the frustum is a rectangle of length 24 cm and width 20 cm. The height of the pyramid *VABCD* is 7 cm.



The volume of the frustum is

A. 70 cm^3.
B. 3290 cm^3.
C. 3948 cm^3.
D. 4410 cm^3.

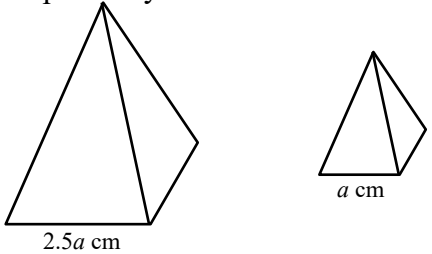
16. The figure shows a right circular cone of slant height 13 cm. If the circumference of the base is 10π cm, the total surface area of the circular cone is



A. 65π cm^2.
B. 90π cm^2.
C. 150π cm^2.
D. 181π cm^2.

17. If the volume of a hemisphere is 486π cm^3, then the diameter of the hemisphere is
A. 7.14 cm, *cor. to 3 sig. fig.*
B. 9 cm.
C. 14.3 cm, *cor. to 3 sig. fig.*
D. 18 cm.

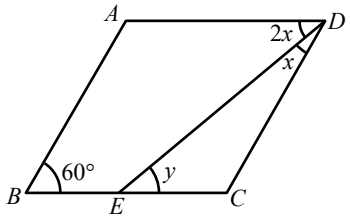
18. In the figure, the lengths of corresponding sides of two similar pyramids are $2.5a$ cm and a cm respectively.



If the volume of the larger pyramid is 1875 cm^3, then the volume of the smaller pyramid is

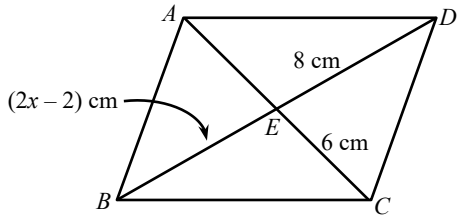
A. 4687.5 cm^3.
B. 750 cm^3.
C. 300 cm^3.
D.120 cm^3.

19. In the figure, $ABCD$ is a parallelogram. E is a point on BC. $\angle ABC = 60°$. Find y.



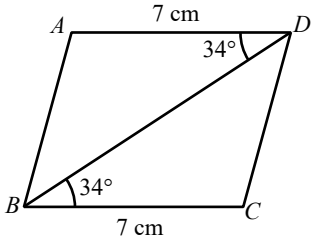
A. 20°
B. 30°
C. 40°
D. 60°

20. In the figure, $ABCD$ is a parallelogram. AC and BD intersect at E. $CE = 6$ cm and $DE = 8$ cm. Find x.



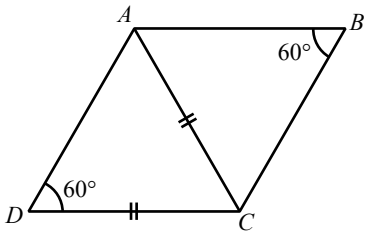
A. 4
B. 5
C. 6
D. 8

21. Refer to the figure.



Which of the following reasons can be used to prove that $ABCD$ is a parallelogram?
A. opp. sides equal
B. opp. $\angle$s equal
C. 2 sides // and equal
D. diags. bisect each other

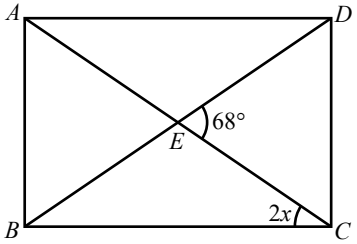
22. In the figure, $ABCD$ is a quadrilateral, where $CD = CA$ and $\angle ABC = \angle ADC = 60°$. Which of the following must be true?



I. $\triangle ADC$ is an equilateral triangle.
II. $ABCD$ is a parallelogram.
III. $\triangle ADC \cong \triangle CBA$

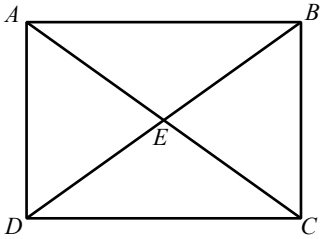
A. I only
B. III only
C. I and II only
D. II and III only

23. In the figure, $ABCD$ is a rectangle. AC and BD intersect at E. $\angle CED = 68°$. Find x.



A. 56°
B. 34°
C. 28°
D. 17°

24. In the figure, $ABCD$ is a rectangle. AC and BD intersect at E.



Which of the following must be true?

I. $\triangle BEC$ is a right-angled triangle.
II. $\triangle CDB$ is a right-angled triangle.
III. $\triangle ADE \cong \triangle BCE$

A. I only
B. II only
C. I and III only
D. II and III only

Section B

25. $ex - ey - ez + fx - fy - fz =$
    A. $(e + f)(x + y + z)$.
    B. $(e + f)(x - y - z)$.
    C. $(e - f)(x + y + z)$.
    D. $(e - f)(x - y - z)$.

26. $12a^2 - 10ab - 42b^2 =$
    A. $3(a - 2b)(4a + 7b)$.
    B. $2(2a - 3b)(3a + 7b)$.
    C. $2(3a - 7b)(2a + 3b)$.
    D. $6(a + 7b)(2a - b)$.

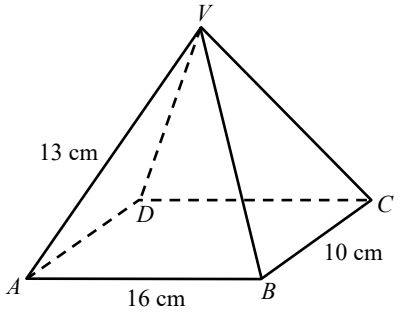
27. $1110000111_2 =$
    A. $2^9 + 2^8 + 2^7 + 7$.
    B. $2^{10} + 2^9 + 2^8 + 7$.
    C. $2^9 + 2^8 + 2^7 + 14$.
    D. $2^{10} + 2^9 + 2^8 + 14$.

28. Two balls are randomly drawn at the same time from four balls numbered 3, 5, 7 and 9 respectively. Find the probability that the difference of the numbers on the balls drawn is 2.
    A. $\frac{1}{6}$
    B. $\frac{3}{8}$
    C. $\frac{1}{2}$
    D. $\frac{2}{3}$

29. Tommy's expenditure decreases by 5% each month from January to April this year. After that, his expenditure increases by 5% per month in the following three months. If his expenditure is \$8000 in January, find his expenditure in July.
    A. \$7540, *cor. to 3 sig. fig.*
    B. \$7920, *cor. to 3 sig. fig.*
    C. \$7940, *cor. to 3 sig. fig.*
    D. \$8000

30. A sum of \$50 000 is deposited in a bank. The interest is compounded half-yearly. After 1 year, an amount of \$51 005 will be got. Find the interest rate per annum.
    A. 1%
    B. 2%
    C. 2.01%
    D. 5.25%

31. The figure shows a right pyramid *VABCD* of slant edge 13 cm. Its base is a rectangle of length 16 cm and width 10 cm.



The total surface area of the pyramid is

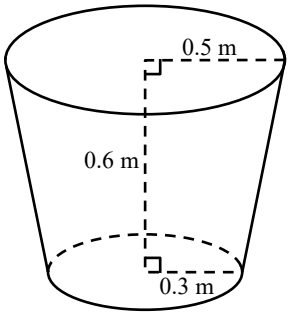
A. $(280+16\sqrt{105})$ cm^2.

B. $(160+32\sqrt{105})$ cm^2.

C. $(120+16\sqrt{105})$ cm^2.

D. 400 cm^2.

32. The figure shows a water bucket in the shape of an inverted right frustum. The radii of the upper base and the lower base are 0.5 m and 0.3 m respectively. The height of the bucket is 0.6 m.



The bucket is fully filled with water. The water is poured into several identical right circular cylindrical containers of base radius 25 cm and height 15 cm. The least number of cylindrical containers needed to carry all the water in the bucket is

A. 9.

B. 10.

C. 11.

D. 12.

33. If the total surface area of a hemisphere is 243π cm^2, then the volume of the hemisphere is

A. 108π cm^2.

B. 243π cm^2.

C. 486π cm^2.

D. 972π cm^2.

34. Consider two similar solids. The ratio of the total surface area of the smaller solid to that of the larger solid is 25 : 36. If the volume of the larger solid is 43.2 cm^3, then the volume of the smaller solid is

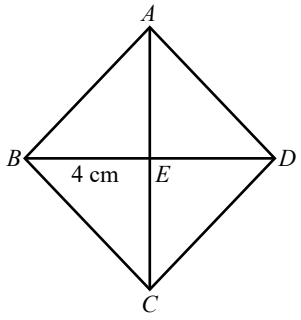
A. 25 cm^3.

B. 30 cm^3.

C. 36 cm^3.

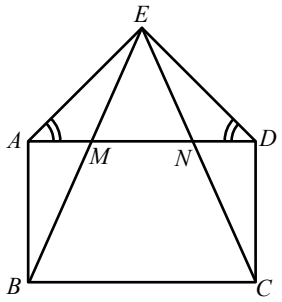
D. 216 cm^3.

35. In the figure, $ABCD$ is a rhombus. AC and BD intersect at E. $BE = 4$ cm. The area of rhombus $ABCD$ is 33.6 cm^2. Find the perimeter of rhombus $ABCD$.



A. 46.4 cm
B. 23.2 cm
C. 16.8 cm
D. 5.8 cm

36. In the figure, $ABCD$ is a rectangle. BE and CE intersect AD at M and N respectively and $\angle EAM = \angle EDN$.



Which of the following must be true?

I. $EM = EN$
II. $\angle BMN = \angle CNM$
III. $\triangle EAB \cong \triangle EDC$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

End of Paper

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